

Summer 2026 STAT 517 Midterm

1. (10 = 4 + 6 Points) Let $X_1, X_2, \dots, X_9$ be a random sample of size 9 from a normal distribution $N(\mu, \sigma^2)$.

- (a) Suppose σ is known. Derive the length of a 95% confidence interval for μ .
- (b) Suppose σ is unknown. Find the expected value of the length of a 95% confidence interval for μ , where S^2 is the sample variance.

Hint: Use the fact that for $W \sim \chi_k^2$, $\mathbb{E}[\sqrt{W}] = \sqrt{2} \frac{\Gamma((k+1)/2)}{\Gamma(k/2)}$.

2. (10 = 5 + 5 Points) Assume that $X_1, \dots, X_m$ is a random sample from $N(\mu, \sigma^2)$ and $Y_1, \dots, Y_n$ is a random sample from $N(0, \sigma^2)$. Further assume that $X_1, \dots, X_m$ are independent of $Y_1, \dots, Y_n$. Let

$$S^2 = \frac{1}{n+m} \left(\sum_{i=1}^n Y_i^2 + \sum_{j=1}^m (X_j - \bar{X})^2 \right), \quad \bar{X} = \frac{1}{m} \sum_{i=1}^m X_i.$$

- (a) Is S^2 an unbiased estimator of σ^2 ? Why? If not, construct an unbiased estimator of σ^2 based on S^2 . Justify your answer.
- (b) Is S^2 a consistent estimator of σ^2 as $n, m \rightarrow \infty$ with $\frac{n}{n+m} \rightarrow \lambda \in (0, 1)$? Why? If not, construct a consistent estimator of σ^2 based on S^2 . Justify your answer.

3. (10 = 5 + 5 Points)

Let $X_1, \dots, X_n$ be a random sample from an Exponential distribution with unknown rate parameter $\theta > 0$, i.e.,

$$f(x; \theta) = \theta e^{-\theta x}, \quad x > 0.$$

Let $p = P(X_i \leq 1) = 1 - e^{-\theta}$. We consider two estimators of p . The first estimator uses the plug-in method with the $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, so $\hat{p} = 1 - e^{-1/\bar{X}_n}$. The second estimator is the empirical proportion: $\tilde{p} = \frac{\#\{i \leq n: X_i \leq 1\}}{n}$.

- (a) Derive the limiting distribution of $\sqrt{n}(\hat{p} - p)$ as $n \rightarrow \infty$.
- (b) Derive the limiting distribution of $\sqrt{n}(\tilde{p} - p)$ as $n \rightarrow \infty$.

4. (10 = 6 + 4 Points) Suppose $Y_1, \dots, Y_n$ are the order statistics of a random sample of size n from a distribution with pdf $f(x; \theta) = \begin{cases} \frac{1}{\theta}, & 0 < x < \theta, \\ 0, & \text{otherwise,} \end{cases}$ where $\theta > 0$. We wish to test the hypothesis $H_0 : \theta = 1$ vs. $H_1 : \theta > 1$ at the $\alpha = 0.05$ level of significance, and we want the power $1 - \beta = 0.80$ when $\theta = 1.1$. We use the following test rule: Reject H_0 if $Y_n \geq c$, where Y_n is the maximum order statistic of the sample.

- (a) What is the smallest sample size (i.e., the smallest n) that will achieve that objective?
- (b) Express the power function for this test.